

PRÁCTICA 10

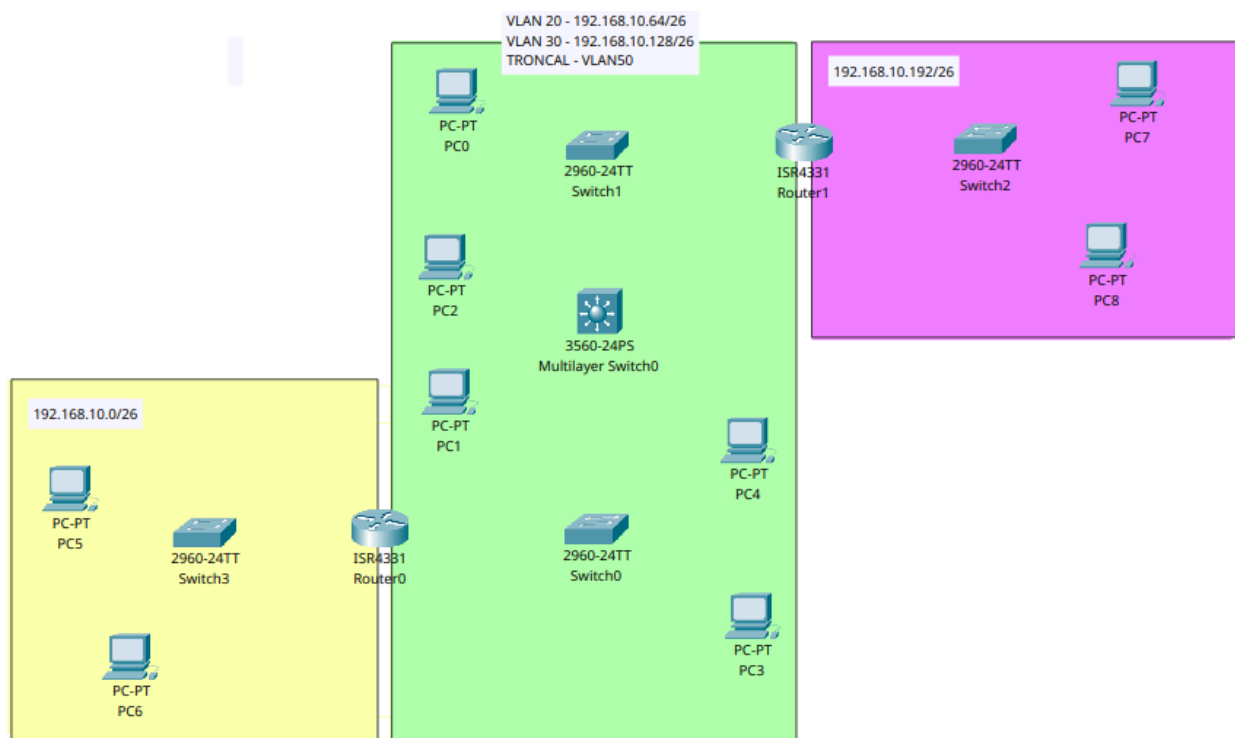
VLANs

1. Diseño y Direccionamiento	3
Esquema de Red Inicial	3
2. Montaje y Escenario Físico	4
Dispositivos Usados	4
Esquema de Red Final	5
Configuración Red Amarilla	6
Configuración Router0	6
Configuración Switch3	7
Configuración PCs	7
Prueba de Conectividad Red Amarilla	8
Ping desde PC5 a PC6 y Router0	8
Ping desde PC6 a PC5 y Router0	8
Configuración Red Rosa	9
Configuración Router1	9
Configuración Switch2	10
Configuración PCs	10
Prueba de Conectividad Red Rosa	11
Ping desde PC7 a PC8 y Router1	11
Ping desde PC8 a PC7 y Router1	11
Configuración Red Verde	12
Configuración Multilayer Switch0	12
Configuración Switch0	14
Configuración Switch1	15
Configuración PCs	16
Prueba de Conectividad Red Verde	17
Pings entre PCs mismas VLANs	17
Pings entre PCs de distintas VLANs	18
Configuración para la interconexión entre redes	19

1. Diseño y Direccionamiento

TABLA DE DIRECCIONAMIENTO	
Red Amarilla	192.168.10.0/26
Red Verde (VLAN 20)	192.168.10.64/26
Red Verde (VLAN 30)	192.168.10.128/26
Red Rosa	192.168.10.192/26

Esquema de Red Inicial



Este es el esquema de red del que disponemos, compuesto de **4 redes principales** y **2 redes de enlace** (enlace entre Router0 y Switch Multicapa, y enlace entre Router1 y Switch Multicapa), posteriormente se conectarán y se configurarán los dispositivos.

2. Montaje y Escenario Físico

Dispositivos Usados

1. Dispositivos de Interconexión (Capa 3 - Red)

- **Routers (ISR4331):** Aparecen dos (Router0 y Router1). Su función es conectar la red amarilla y la rosa con el resto del sistema. En ellos se configurará el DHCP para sus respectivas redes.
- **Switch Multicapa (3560-24PS):** Es el dispositivo central de la red verde. A diferencia de un switch normal, este puede enrutar tráfico entre las VLANs (Inter-VLAN routing) y actuar como servidor DHCP.

2. Dispositivos de Distribución y Acceso (Capa 2 - Enlace)

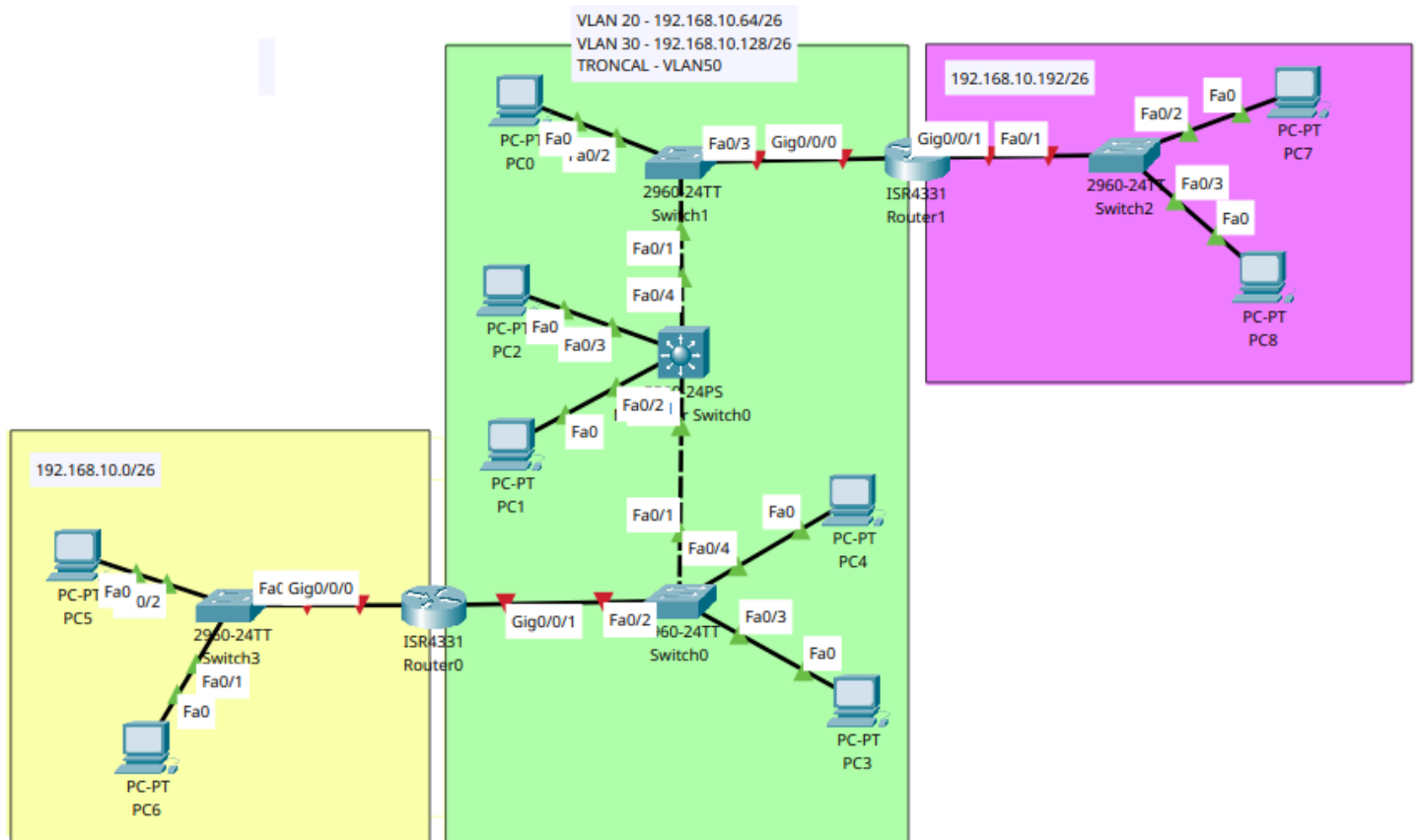
- **Switches de Capa 2 (2960-24TT):** Hay tres en total (Switch0, Switch1 y Switch3).
 - Switch1 y Switch0: Se encargan de la red verde, distribuyendo los puertos hacia las VLANs 20 y 30.
 - Switch3 y Switch2: Simplemente extienden la red en las zonas amarilla y rosa (donde no hay VLANs según las instrucciones).

3. Dispositivos Finales (Hosts)

PCs Genéricos (PC-PT): Hay un total de 9 ordenadores distribuidos así:

- **Red Amarilla:** 2 PCs (PC5, PC6).
- **Red Verde:** 5 PCs (PC0, PC1 y PC2 para la VLAN 20, PC3, PC4 para la VLAN 30).
- **Red Rosa:** 2 PCs (PC7, PC8).

Esquema de Red Final



Configuración Red Amarilla

Configuración Router0

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0

Router(config)#interface Gig0/0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool RED_AMARILLA
Router(dhcp-config)#network 192.168.10.0 255.255.255.192
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit

Router0#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En su configuración incluiremos la asignación de la IP en la interfaz **Gig/0/0/0** con **ip address** para establecer la puerta de enlace, y el comando **no shutdown** para habilitar el paso de datos. Además, ejecutaremos **ip dhcp excluded-address** para reservar las 10 primeras IPs, seguido de **ip dhcp pool** y **network** para crear el servicio que entrega automáticamente direcciones a los clientes de la red.

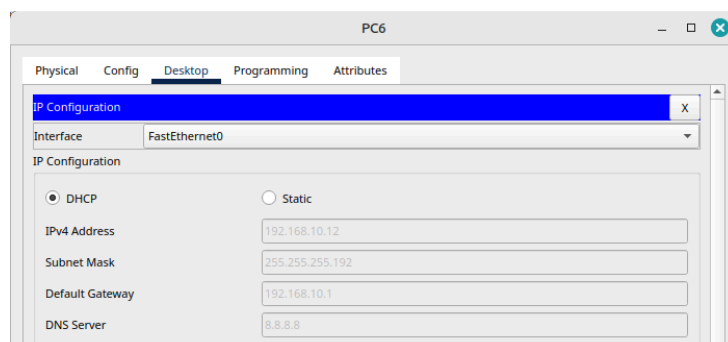
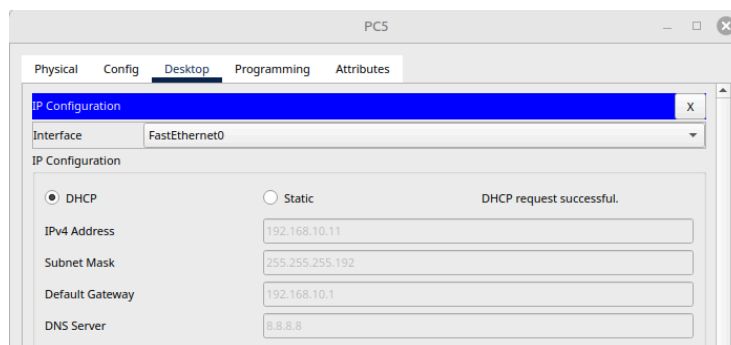
Configuración Switch3

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch3
Switch(config)#exit

Switch3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En su configuración nos limitaremos a la gestión básica del dispositivo y verificación de puertos, ya que al no existir VLANs, el switch funciona de manera predeterminada en modo de acceso. Su propósito es simplemente conmutar el tráfico de los PC5 y PC6 hacia el router, por lo que utilizaremos comandos como **hostname** para identificación y **copy running-config startup-config** para asegurar que el estado operativo se mantenga tras un reinicio.

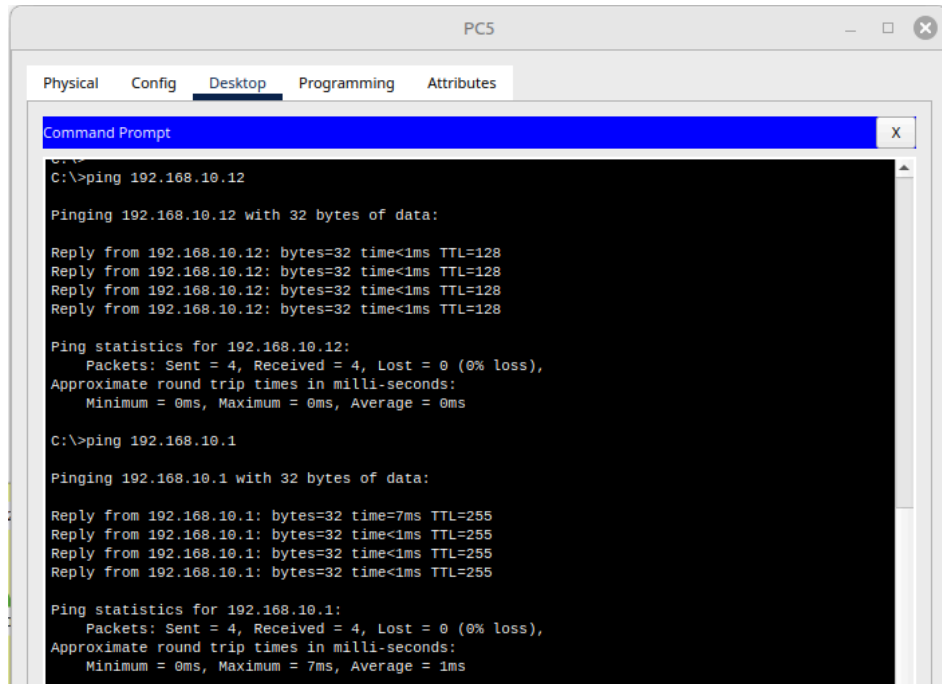
Configuración PCs



Para configurar los PCs simplemente desde el apartado **Desktop** → **IP Configuration** marcaremos la opción DHCP en la sección de **IP Configuration**, asignando las IPs automáticamente debido a DHCP.

Prueba de Conectividad Red Amarilla

Ping desde PC5 a PC6 y Router0



The screenshot shows a Cisco Packet Tracer window for PC5. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The user has entered two ping commands. The first ping is to 192.168.10.12, which returns four successful replies with 0ms round trip times. The second ping is to 192.168.10.1, which also returns four successful replies with 7ms round trip times.

```
C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

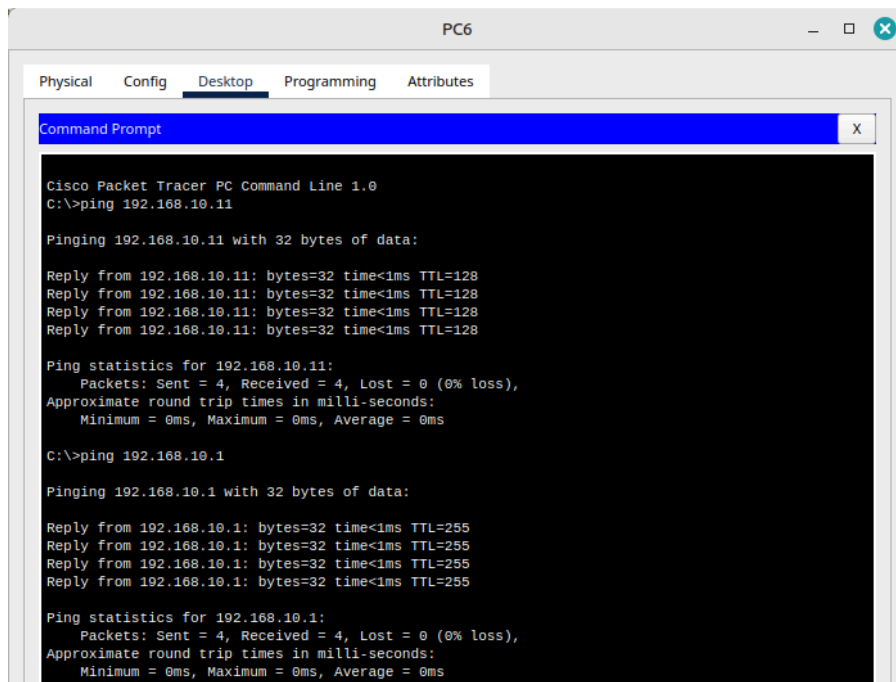
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=7ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms
```

Ping desde PC6 a PC5 y Router0



The screenshot shows a Cisco Packet Tracer window for PC6. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The user has entered two ping commands. The first ping is to 192.168.10.11, which returns four successful replies with 0ms round trip times. The second ping is to 192.168.10.1, which also returns four successful replies with 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```


Configuración Red Rosa

Configuración Router1

```
Router>enable
Router#configure terminal
Router(config)#hostname Router1

Router1(config)#interface Gig0/0/1
Router1(config-if)#ip address 192.168.10.193 255.255.255.192
Router1(config-if)#no shutdown
Router1(config-if)#exit

Router1(config)#ip dhcp excluded-address 192.168.10.193 192.168.10.202
Router1(config)#ip dhcp pool RED_ROSA
Router1(dhcp-config)#network 192.168.10.192 255.255.255.192
Router1(dhcp-config)#default-router 192.168.10.193
Router1(dhcp-config)#exit
Router1(config)#exit

Router1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En su configuración incluiremos la asignación de la IP en la interfaz **Gig/0/0/1** con **ip address** para establecer la puerta de enlace, y el comando **no shutdown** para habilitar el paso de datos. Además, ejecutaremos **ip dhcp excluded-address** para reservar las 10 primeras IPs, seguido de **ip dhcp pool** y **network** para crear el servicio que entrega automáticamente direcciones a los clientes de la red.

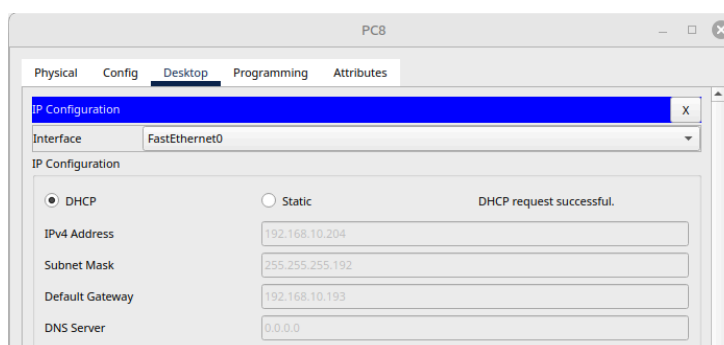
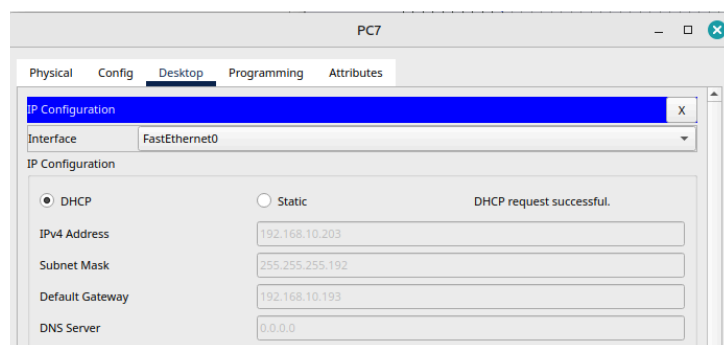
Configuración Switch2

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch2
Switch2(config)#exit

Switch2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En su configuración nos limitaremos a la gestión básica del dispositivo y verificación de puertos, ya que al no existir VLANs, el switch funciona de manera predeterminada en modo de acceso. Su propósito es simplemente conmutar el tráfico de los PC5 y PC6 hacia el router, por lo que utilizaremos comandos como **hostname** para identificación y **copy running-config startup-config** para asegurar que el estado operativo se mantenga tras un reinicio.

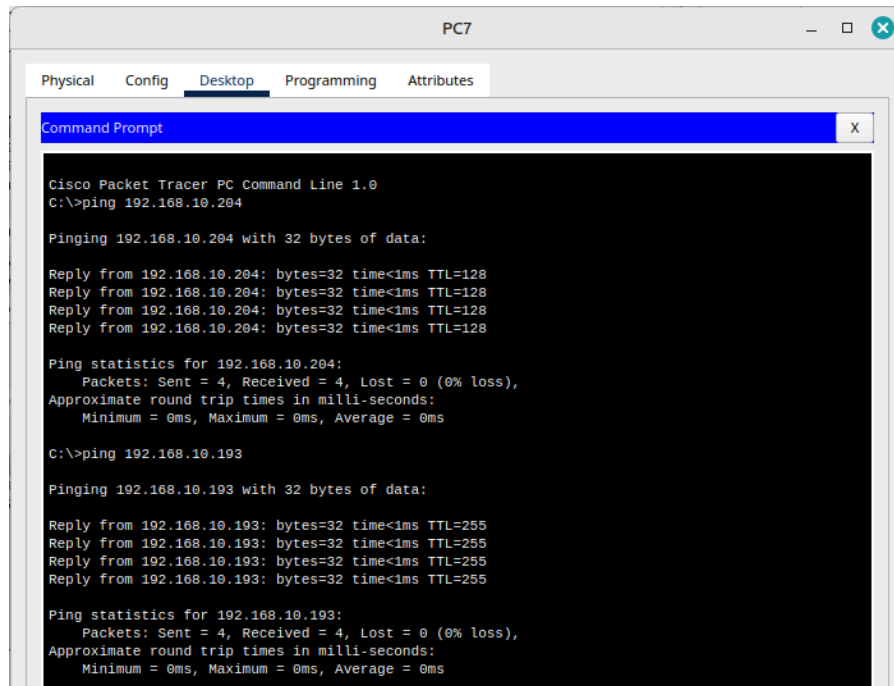
Configuración PCs



Para configurar los PCs simplemente desde el apartado **Desktop** → **IP Configuration** marcaremos la opción DHCP en la sección de **IP Configuration**, asignando las IPs automáticamente debido a DHCP.

Prueba de Conectividad Red Rosa

Ping desde PC7 a PC8 y Router1



The screenshot shows the Cisco Packet Tracer interface for PC7. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the results of two ping commands. The first command is 'ping 192.168.10.204', which returns four successful replies with a TTL of 128. The second command is 'ping 192.168.10.193', which returns four successful replies with a TTL of 255. Both commands show 0% packet loss and 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.204

Pinging 192.168.10.204 with 32 bytes of data:

Reply from 192.168.10.204: bytes=32 time<1ms TTL=128
Reply from 192.168.10.204: bytes=32 time<1ms TTL=128
Reply from 192.168.10.204: bytes=32 time<1ms TTL=128
Reply from 192.168.10.204: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.204:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

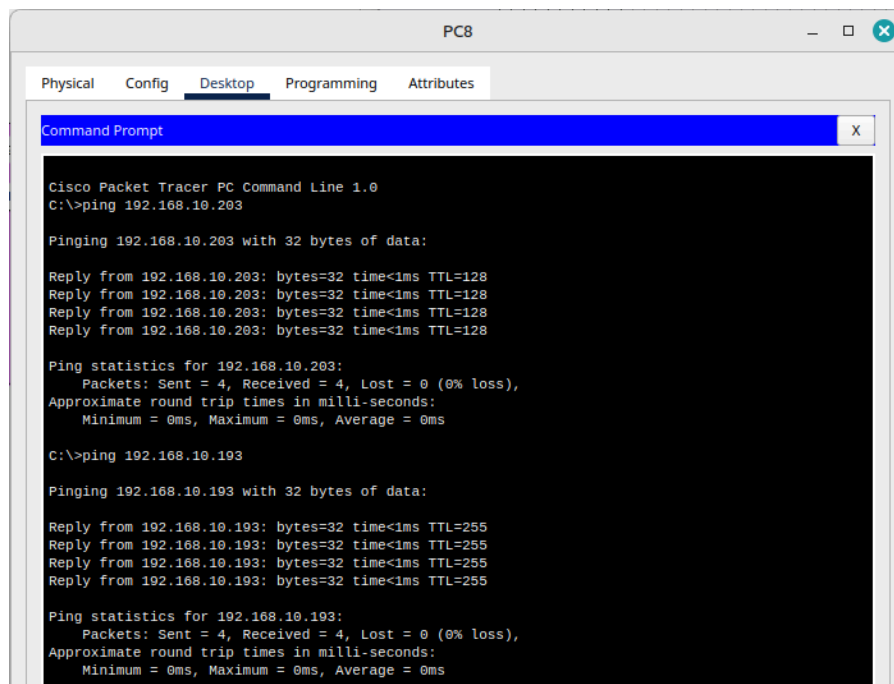
C:\>ping 192.168.10.193

Pinging 192.168.10.193 with 32 bytes of data:

Reply from 192.168.10.193: bytes=32 time<1ms TTL=255
Reply from 192.168.10.193: bytes=32 time<1ms TTL=255
Reply from 192.168.10.193: bytes=32 time<1ms TTL=255
Reply from 192.168.10.193: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping desde PC8 a PC7 y Router1



The screenshot shows the Cisco Packet Tracer interface for PC8. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the results of two ping commands. The first command is 'ping 192.168.10.203', which returns four successful replies with a TTL of 128. The second command is 'ping 192.168.10.193', which returns four successful replies with a TTL of 255. Both commands show 0% packet loss and 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.203

Pinging 192.168.10.203 with 32 bytes of data:

Reply from 192.168.10.203: bytes=32 time<1ms TTL=128
Reply from 192.168.10.203: bytes=32 time<1ms TTL=128
Reply from 192.168.10.203: bytes=32 time<1ms TTL=128
Reply from 192.168.10.203: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.203:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.193

Pinging 192.168.10.193 with 32 bytes of data:

Reply from 192.168.10.193: bytes=32 time<1ms TTL=255
Reply from 192.168.10.193: bytes=32 time<1ms TTL=255
Reply from 192.168.10.193: bytes=32 time<1ms TTL=255
Reply from 192.168.10.193: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Configuración Red Verde

Configuración Multilayer Switch0

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname MLS_Verde

MLS_Verde(config)#vlan 20
MLS_Verde(config-vlan)#name VLAN20
MLS_Verde(config-vlan)#exit
MLS_Verde(config)#vlan 30
MLS_Verde(config-vlan)#name VLAN30
MLS_Verde(config-vlan)#exit
Switch0(config)#vlan 50
Switch0(config-vlan)#name TRONCAL
Switch0(config-vlan)#exit

MLS_Verde(config)#interface vlan 20
MLS_Verde(config-if)#ip address 192.168.10.65 255.255.255.192
MLS_Verde(config-if)#no shutdown
MLS_Verde(config-if)#exit
MLS_Verde(config)#interface vlan 30
MLS_Verde(config-if)#ip address 192.168.10.129 255.255.255.192
MLS_Verde(config-if)#no shutdown
MLS_Verde(config-if)#exit

MLS_Verde(config)#ip routing

MLS_Verde(config)#ip dhcp excluded-address 192.168.10.65 192.168.10.75
MLS_Verde(config)#ip dhcp excluded-address 192.168.10.129 192.168.10.139

MLS_Verde(config)#ip dhcp pool VLAN20
MLS_Verde(dhcp-config)#network 192.168.10.64 255.255.255.192
MLS_Verde(dhcp-config)#default-router 192.168.10.65
MLS_Verde(dhcp-config)#dns-server 8.8.8.8
MLS_Verde(dhcp-config)#exit
MLS_Verde(config)#ip dhcp pool VLAN30
MLS_Verde(dhcp-config)#network 192.168.10.128 255.255.255.192
MLS_Verde(dhcp-config)#default-router 192.168.10.129
MLS_Verde(dhcp-config)#dns-server 8.8.8.8
MLS_Verde(dhcp-config)#exit
```

```
MLS_Verde(config)#interface Fa0/2
MLS_Verde(config-if)#switchport mode access
MLS_Verde(config-if)#switchport access vlan 20
MLS_Verde(config-if)#spanning-tree portfast
MLS_Verde(config-if)#exit
MLS_Verde(config)#interface Fa0/3
MLS_Verde(config-if)#switchport mode access
MLS_Verde(config-if)#switchport access vlan 20
MLS_Verde(config-if)#spanning-tree portfast
MLS_Verde(config-if)#exit
MLS_Verde(config)#interface Fa0/1
MLS_Verde(config-if)#switchport trunk encapsulation dot1q
MLS_Verde(config-if)#switchport mode trunk
MLS_Verde(config-if)#switchport trunk allowed vlan 20,30,50
MLS_Verde(config-if)#exit
MLS_Verde(config)#interface Fa0/4
MLS_Verde(config-if)#switchport trunk encapsulation dot1q
MLS_Verde(config-if)#switchport mode trunk
MLS_Verde(config-if)#switchport trunk allowed vlan 20,30,50
MLS_Verde(config-if)#exit
MLS_Verde(config)#exit

MLS_Verde#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En la configuración del **Switch Multicapa (MLS_Verde)** se ha establecido el núcleo inteligente de la red verde para gestionar el tráfico de forma centralizada. En su configuración incluiremos la activación del enrutamiento IP y la creación de las **VLANs 20, 30 y 50**, definiendo puertos de acceso para equipos finales y puertos **troncales** con encapsulación **dot1q** para la comunicación con otros switches. Además, se han implementado servidores **DHCP** para ambas subredes con exclusiones de seguridad, lo que permite que el dispositivo asigna automáticamente direcciones IP a los usuarios mientras actúa como la puerta de enlace (Gateway) que permite la comunicación entre las diferentes VLANs de la organización.

Configuración Switch0

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch0

Switch0(config)#vlan 30
Switch0(config-vlan)#name VLAN30
Switch0(config-vlan)#exit
Switch0(config)#vlan 50
Switch0(config-vlan)#name TRONCAL
Switch0(config-vlan)#exit

Switch0(config)#interface Fa0/3
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 30
Switch0(config-if)#spanning-tree portfast
Switch0(config-if)#exit
Switch0(config)#interface Fa0/4
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 30
Switch0(config-if)#spanning-tree portfast
Switch0(config-if)#interface Fa0/1
Switch0(config-if)#switchport mode trunk
Switch0(config-if)#switchport trunk allowed vlan 20,30,50
Switch0(config-if)#exit
Switch0(config)#exit

Switch0#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En el **Switch0 (Capa 2)** se ha configurado la segmentación local de la red para organizar el tráfico de los dispositivos finales. En su configuración incluiremos la creación de las **VLANs 20, 30 y 50**, definiendo qué puertos actuarán como puntos de acceso para los usuarios y cuál servirá como enlace principal. Específicamente, se han asignado los puertos **Fa0/3 y Fa0/4** a la VLAN 30 para conectar los equipos de esa subred, mientras que el puerto **Fa0/1** se ha establecido en **modo troncal (trunk)** para permitir que el tráfico etiquetado de todas las VLANs viaje de forma organizada hacia el Switch Multicapa.

Configuración Switch1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch1

Switch1(config)#vlan 20
Switch1(config-vlan)#name VLAN20
Switch1(config-vlan)#exit
Switch1(config)#vlan 50
Switch1(config-vlan)#name TRONCAL
Switch1(config-vlan)#exit

Switch1(config)#interface Fa0/2
Switch1(config-if)#switchport mode access
Switch1(config-if)#switchport access vlan 20
Switch1(config-if)#spanning-tree portfast
Switch1(config-if)#exit
Switch1(config)#interface Fa0/1
Switch1(config-if)#switchport mode trunk
Switch1(config-if)#switchport trunk allowed vlan 20,30,50
Switch1(config-if)#exit
Switch1(config)#exit

Switch1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

En el **Switch1 (Capa 2)** se ha completado la estructura de la red verde mediante la configuración de los dominios de colisión locales. En su configuración incluiremos la creación de las **VLANs 20, 30 y 50**, junto con la asignación del puerto **Fa0/2** a la VLAN 20 para integrar los equipos correspondientes a este segmento. Al igual que en el resto de la topología, se ha establecido el puerto **Fa0/1** en **modo troncal (trunk)**, permitiendo que el tráfico de todas las VLANs configuradas se comuniquen con el Switch Multicapa para su correcto enrutamiento y salida hacia el resto de la red.

Configuración PCs

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /renew

IP Address.....: 192.168.10.76
Subnet Mask.....: 255.255.255.192
Default Gateway...: 192.168.10.65
DNS Server.....: 8.8.8.8
```

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>ipconfig /renew

IP Address.....: 192.168.10.77
Subnet Mask.....: 255.255.255.192
Default Gateway...: 192.168.10.65
DNS Server.....: 8.8.8.8
```

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /renew

IP Address.....: 192.168.10.78
Subnet Mask.....: 255.255.255.192
Default Gateway...: 192.168.10.65
DNS Server.....: 8.8.8.8
```

PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>ipconfig /renew

IP Address.....: 192.168.10.140
Subnet Mask.....: 255.255.255.192
Default Gateway...: 192.168.10.129
DNS Server.....: 8.8.8.8
```

PC4

Physical Config Desktop Programming Attributes

Command Prompt

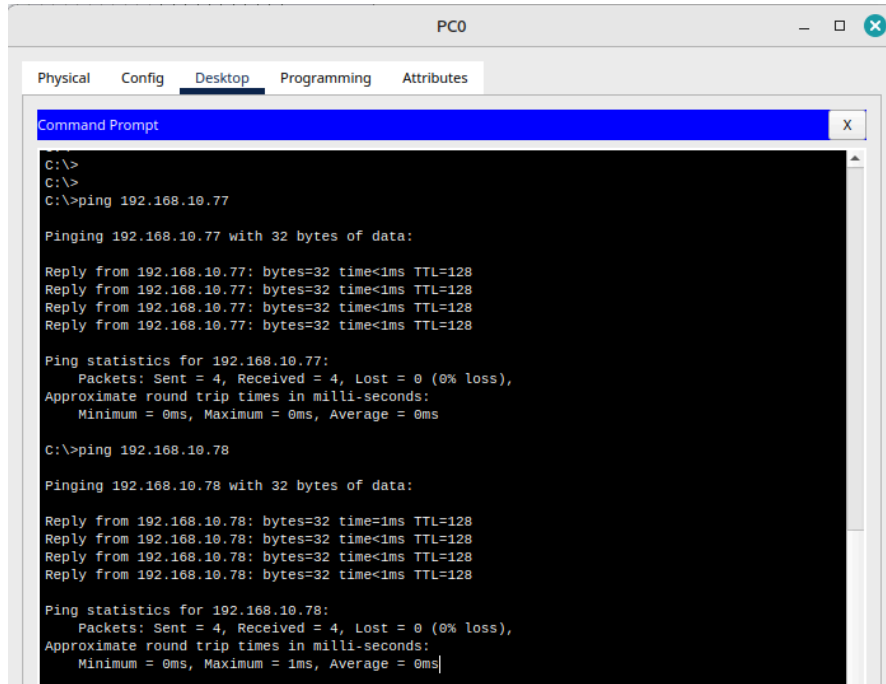
```
C:\>
C:\>
C:\>ipconfig /renew

IP Address.....: 192.168.10.141
Subnet Mask.....: 255.255.255.192
Default Gateway...: 192.168.10.129
DNS Server.....: 8.8.8.8
```

Además de la opción anteriormente vista, también se pueden configurar los PCs desde el apartado **Desktop** → **Command Prompt** escribiendo el comando **ipconfig /renew**, asignando las IPs automáticamente debido a DHCP.

Prueba de Conectividad Red Verde

Pings entre PCs mismas VLANs



PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>ping 192.168.10.77

Pinging 192.168.10.77 with 32 bytes of data:

Reply from 192.168.10.77: bytes=32 time<1ms TTL=128
Reply from 192.168.10.77: bytes=32 time<1ms TTL=128
Reply from 192.168.10.77: bytes=32 time<1ms TTL=128
Reply from 192.168.10.77: bytes=32 time<1ms TTL=128

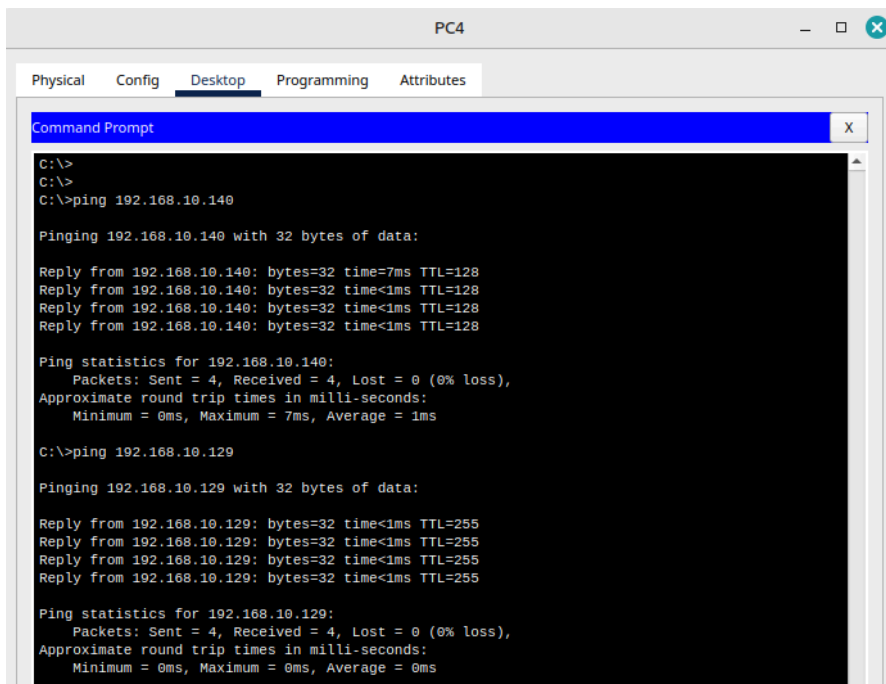
Ping statistics for 192.168.10.77:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.78

Pinging 192.168.10.78 with 32 bytes of data:

Reply from 192.168.10.78: bytes=32 time=1ms TTL=128
Reply from 192.168.10.78: bytes=32 time<1ms TTL=128
Reply from 192.168.10.78: bytes=32 time<1ms TTL=128
Reply from 192.168.10.78: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms|
```



PC4

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>ping 192.168.10.140

Pinging 192.168.10.140 with 32 bytes of data:

Reply from 192.168.10.140: bytes=32 time=7ms TTL=128
Reply from 192.168.10.140: bytes=32 time<1ms TTL=128
Reply from 192.168.10.140: bytes=32 time<1ms TTL=128
Reply from 192.168.10.140: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.140:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

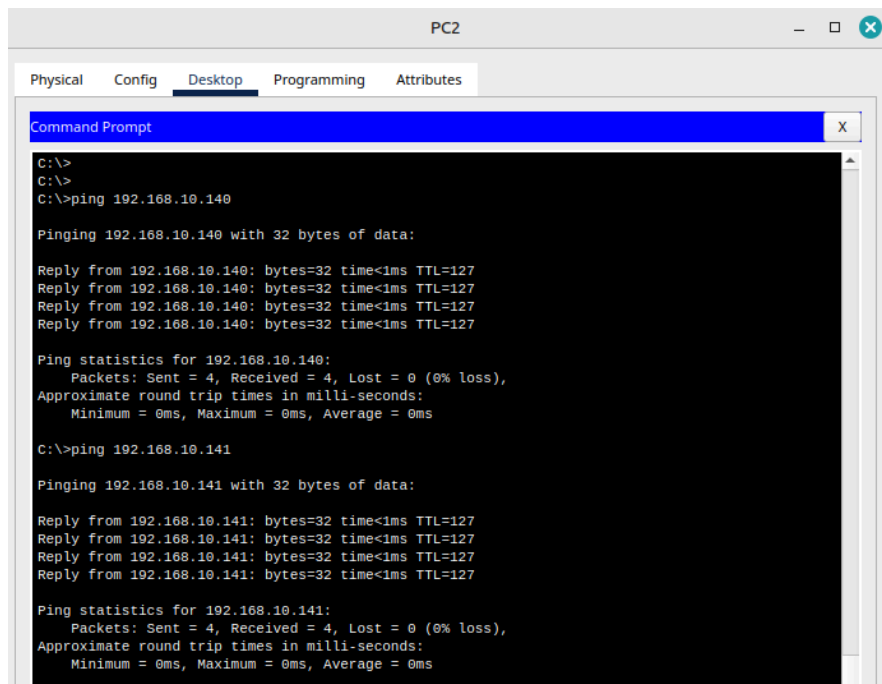
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Pings entre PCs de distintas VLANs



PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>ping 192.168.10.140

Pinging 192.168.10.140 with 32 bytes of data:

Reply from 192.168.10.140: bytes=32 time<1ms TTL=127
Reply from 192.168.10.140: bytes=32 time<1ms TTL=127
Reply from 192.168.10.140: bytes=32 time<1ms TTL=127
Reply from 192.168.10.140: bytes=32 time<1ms TTL=127

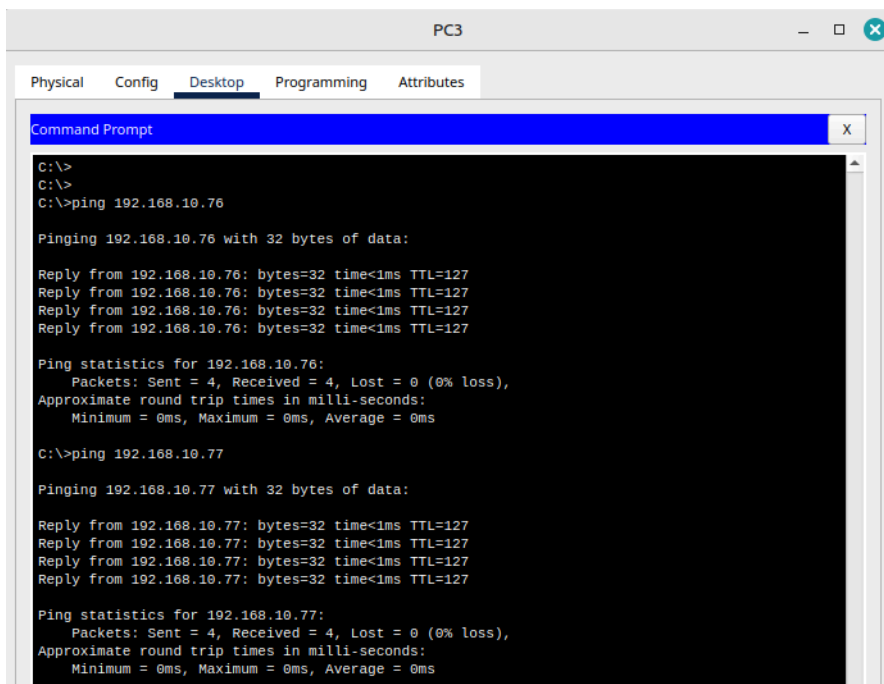
Ping statistics for 192.168.10.140:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.141

Pinging 192.168.10.141 with 32 bytes of data:

Reply from 192.168.10.141: bytes=32 time<1ms TTL=127
Reply from 192.168.10.141: bytes=32 time<1ms TTL=127
Reply from 192.168.10.141: bytes=32 time<1ms TTL=127
Reply from 192.168.10.141: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.141:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>ping 192.168.10.76

Pinging 192.168.10.76 with 32 bytes of data:

Reply from 192.168.10.76: bytes=32 time<1ms TTL=127
Reply from 192.168.10.76: bytes=32 time<1ms TTL=127
Reply from 192.168.10.76: bytes=32 time<1ms TTL=127
Reply from 192.168.10.76: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.76:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.77

Pinging 192.168.10.77 with 32 bytes of data:

Reply from 192.168.10.77: bytes=32 time<1ms TTL=127
Reply from 192.168.10.77: bytes=32 time<1ms TTL=127
Reply from 192.168.10.77: bytes=32 time<1ms TTL=127
Reply from 192.168.10.77: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.77:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Configuración para la interconexión entre redes

Para que los ordenadores de todas las redes puedan comunicarse entre sí, se ha configurado el **enrutamiento estático** en los dispositivos principales. En los **Routers 0 y 1**, se han creado caminos lógicos (subinterfaces) que permiten "entender" el tráfico de las VLANs de la red verde. Por su parte, el **Switch Multicapa** actúa como el núcleo central, encargándose de mover los datos entre las diferentes VLANs y de enviar hacia los routers externos todo el tráfico que no pertenece a la red verde. En resumen, se han establecido "puentes virtuales" y reglas de dirección en cada equipo para que cualquier paquete de datos sepa exactamente qué saltos debe dar para llegar desde su origen hasta cualquier otro punto de la topología.

Configuración Router0

```
Router0>enable
Router0#configure terminal

Router0(config)#interface Gig0/0/1
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface Gig0/0/1.20
Router0(config)#ip address 192.168.10.66 255.255.255.192
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface Gig0/0/1.30
Router0(config)#ip address 192.168.10.130 255.255.255.192
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0(config)#ip route 192.168.10.64 255.255.255.192 192.168.10.126
Router0(config)#ip route 192.168.10.128 255.255.255.192 192.168.10.126
Router0(config)#ip route 192.168.10.192 255.255.255.192 192.168.10.193
Router0(config)#exit

Router0#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Configuración Router1

```
Router1>enable
Router1#configure terminal

Router1(config)#interface Gig0/0/0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface Gig0/0/0.20
Router1(config)#ip address 192.168.10.65 255.255.255.192
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface Gig0/0/0.30
Router1(config)#ip address 192.168.10.129 255.255.255.192
Router1(config-if)#no shutdown
Router1(config-if)#exit

Router1(config)#ip route 192.168.10.0 255.255.255.192 192.168.10.126
Router1(config)#ip route 192.168.10.64 255.255.255.192 192.168.10.126
Router1(config)#ip route 192.168.10.128 255.255.255.192 192.168.10.66
Router1(config)#exit

Router1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Configuración Multilayer Switch0

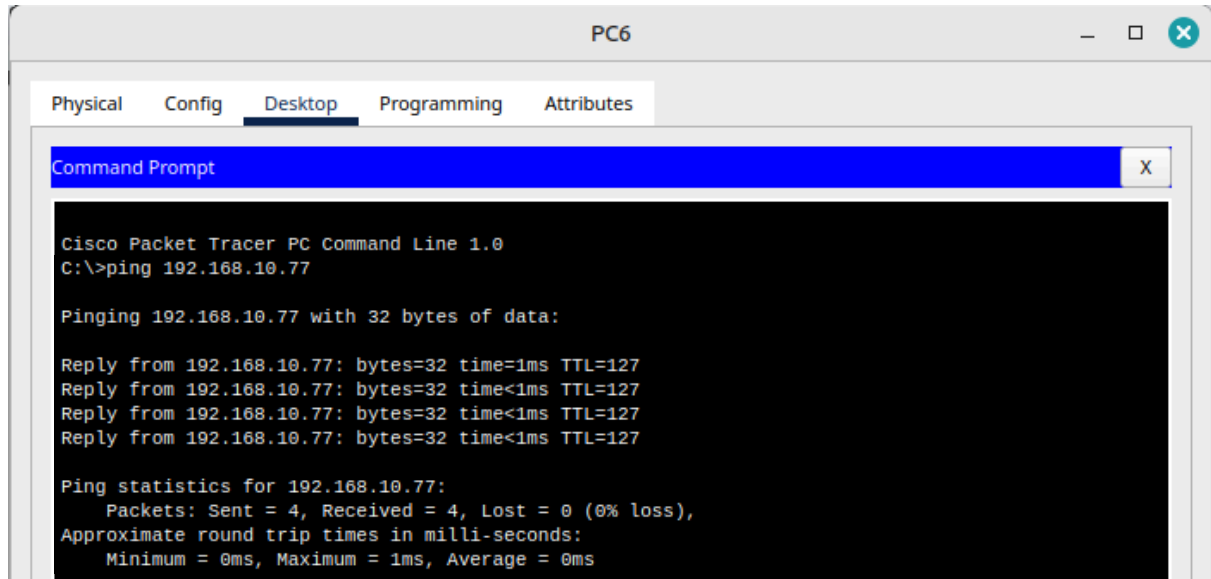
```
MLS_Verde>enable
MLS_Verde#configure terminal

MLS_Verde(config)#ip route 192.168.10.0 255.255.255.192 192.168.10.65
MLS_Verde(config)#ip route 192.168.10.0 255.255.255.192 192.168.10.66
MLS_Verde(config)#exit

MLS_Verde#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Prueba de conectividad entre las redes

Ping entre la red amarilla y la red verde



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC6. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and the Command Prompt is open. The command prompt shows the following output:

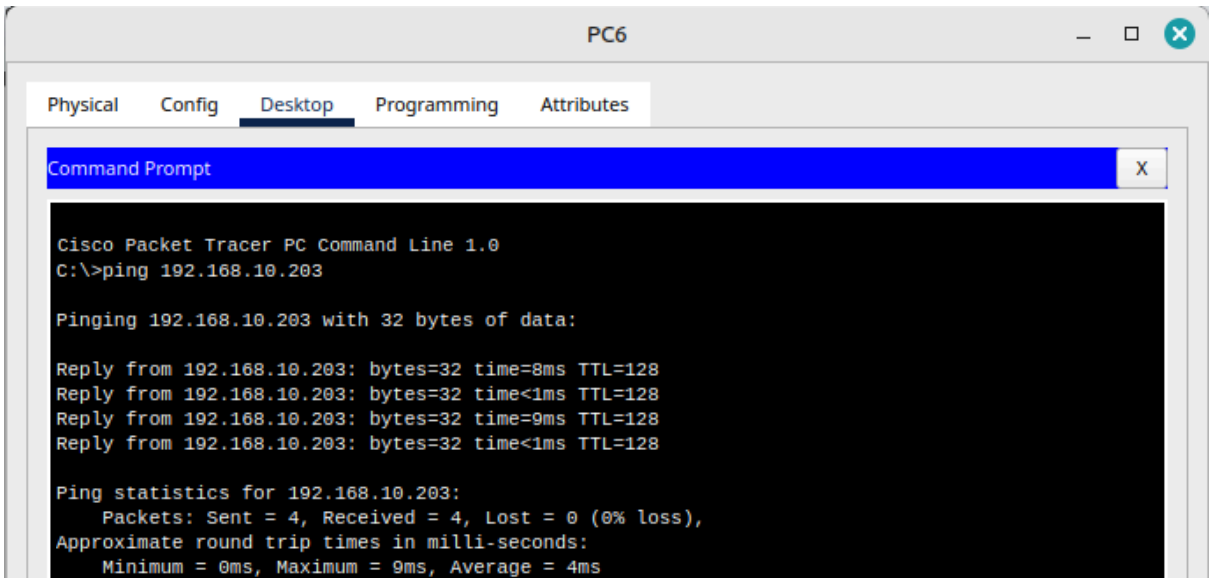
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.77

Pinging 192.168.10.77 with 32 bytes of data:

Reply from 192.168.10.77: bytes=32 time=1ms TTL=127
Reply from 192.168.10.77: bytes=32 time<1ms TTL=127
Reply from 192.168.10.77: bytes=32 time<1ms TTL=127
Reply from 192.168.10.77: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.77:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping entre la red amarilla y la red rosa



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC6. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and the Command Prompt is open. The command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.203

Pinging 192.168.10.203 with 32 bytes of data:

Reply from 192.168.10.203: bytes=32 time=8ms TTL=128
Reply from 192.168.10.203: bytes=32 time<1ms TTL=128
Reply from 192.168.10.203: bytes=32 time=9ms TTL=128
Reply from 192.168.10.203: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.203:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 4ms
```

Ping entre la red verde y la red rosa

